

A fast pre-heating method for lithium-ion batteries by wireless energy transfer at low temperatures

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ABSTRACT

Lithium-ion batteries (LIBs) are required to be preheated at low temperatures in advance to ensure their normal operation. Alternating current (AC) heating is an effective heating method with little damage to the batteries. This study develops a new method to obtain onboard AC power by improving the wireless charging system based on the inductance and double capacitances-series compensation topology, which heats the battery by high-frequency AC on the secondary side of the wireless charging system. Heating experiments of a single cell under different frequencies, states of charge and current amplitudes and the battery packs with different numbers of battery cells in series were conducted. Experimental results indicate that this method can heat the battery from -20°C to 0°C in 142s at an root mean square value of 3C and an average heating rate of about $8.5^{\circ}\text{C}/\text{min}$ with 85 kHz AC. The proposed heating system can provide fast heating of LIBs for electric vehicles.

1. Introduction

Electric vehicles (EVs) play a critical role in revolutionizing the transportation and energy sectors. Owing to the advantages of excellent security and long cycle life, lithium-ion batteries (LIBs) are dominant power sources in EVs [1]. However, their performance is negatively affected by low temperatures [2]. At low temperatures, charging and discharging LIBs become difficult due to the decrease in electrochemical reaction rates [3,4]. Furthermore, charging LIBs at low temperatures may have the risk of lithium plating, which can shorten battery life and give rise to thermal runaway [5]. Preheating LIBs in cold climates is one of the key measures to address these issues.

1.1. Literature review

Many efforts have been made to preheat LIBs. The heating methods can be generally categorized into two groups, namely external heating [6,7] and internal heating [8,9]. Guo et al. [6] proposed a battery thermal management system to use refrigerant to directly heat and cool the battery without auxiliary devices. He et al. [7] developed a method for heating the battery pack using a heating sheet and phase change material. The phase change material is used to balance the temperature

of the battery pack while the heating sheet is used to heat the battery pack. Oehl et al. [8] proposed a high-frequency AC self-heating system that utilizes battery energy to achieve rapid heating. The system is composed of only two semiconductor switches and one capacitor which can be easily implemented while saving cost and space. Ruan et al. [9] developed a DC internal heating strategy based on a simple fade model to capture battery capacity loss, which can quickly heat the battery without significantly reducing battery life. External heating strategies heat LIBs through external sources, such as heat pumps, air, liquids and phase change materials (PCM) [10]. They are safe and easy to accomplish in practical applications but have low heating efficiency and uneven heating. In contrast, internal heating methods heat LIBs through a certain excitation, which can overcome the problem of uneven heating with a satisfying heating rate and efficiency [11]. For example, Wang et al. [12] proposed a self-heating structure by insetting a nickel sheet inside a battery for heat generation. This method can heat the battery from -30°C to 0°C within 30 s at the cost of 5.5% of battery capacity. Qu et al. [13] developed a pulsed self-heating method for battery heating, which can heat the battery from -10°C to 10°C within 175s while DC heating takes 280s to produce the same effect. Ji et al. [14] proposed the mutual pulse heating method, which can heat the battery by adding DC-DC between two groups of batteries to generate mutual

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pulses. This method can control the heating rate by adjusting the pulse amplitude and time.

For DC heating method, the current direction of DC is unchanged during heating, so the amplitude of DC must be limited in order to reduce the damage to the battery, which will result in low heating efficiency [14]. Alternating current (AC) excitation is also one of the effective internal heating methods to preheat a battery [15,16] through heat generation of battery internal impedance. Compared with DC heating, AC heating has small current limits in magnitude and duration to avoid lithium deposition. Furthermore, a high-frequency AC excitation means that charging and discharging durations are very short which can avoid the change of battery state of charge (SOC) and further prevent the occurrence of lithium deposition reactions and the loss of lithium capacity [17–19]. Inspired by this, Xiong et al. [20] proposed an echelon internal heating method based on AC and fabricated a three-electrode battery for verification. The results show that the battery can be preheated from -20°C to 0°C in 5 min without affecting battery health. But this method depends on expensive power supply equipment. Dai et al. [21] analyzed the effect of different AC frequencies, amplitudes and voltage limits on battery degradation while heating the battery. They found that the battery can be uniformly preheated by the AC excitation with the low frequency (0.5 Hz) and there is no irreversible damage to the battery health under the normal voltage protection limitation. However, this research only focuses on low-frequency AC heating. Moreover, Jiang et al. [22] explored the effect of AC frequency on polarization voltage and irreversible heat. A simplified electro-thermal coupling model was proposed for low temperatures. The results show that current amplitude has a greater effect on the battery temperature rise rate than the frequency. However, this is only for the exploration of heating strategy, which still did not solve the problem of application in the vehicle. Although AC heating methods have been experimentally validated to be effective, an AC power source for heating batteries is not readily available in EVs, which limits AC heating applications. Commercial AC power sources like Kikusui bipolar supply are costly and cumbersome [23]. However, this is only for the exploration of heating strategy, which still did not solve the problem of application in the vehicle. Shang et al. [24] designed an automotive on-board AC heater to preheat battery. Experiments showed that this onboard AC heater can preheat the battery from -20°C to 0°C within 5.9 min by consuming approximately 5% of the battery energy. Nevertheless, it consumes battery energy and faces the risk of failure at low battery SOC.

The above analysis implies that AC excitation generation is a limiting factor for the development of AC heating methods in EV applications. A promising solution is to reuse the existing AC power source in EVs. For EV wireless charging systems, scholars often focus on charging efficiency or system stability while ignoring the usefulness of its AC power supply [24–26]. Yuan et al. [27] proposed an inductance and double capacitances-series (LCC-S) compensation topology to improve the stability of the wireless energy transfer system and further designed a compact receiver [28] to improve fault tolerance. However, few research has attempted to reuse the wireless charging systems in EVs for battery heating.

1.2. Contributions of this study

In this study, we design a wireless charging system based on LCC-S compensation topology which can generate AC for battery heating. The proposed method has the following advantages:

- 1) New configuration for integration and device reuse. The AC power is drawn from the secondary side of the wireless charging system without adding extra hardware, which has a simple structure with low cost.
- 2) High and adjustable heating rate. The designed structure can generate an 85 kHz sinusoidal AC on the secondary side with the adjustable heating current through changing the input voltage.

Results show that the battery cell can be heated from -20°C to 0°C in 142s with an average heating rate of about $8.4^{\circ}\text{C}/\text{min}$ for the given heating current with the root mean square (RMS) value of 7800 mA (i.e. 3C with respect to the capacity of the battery cell).

- 3) Ease of extension to battery packs. The proposed configuration can be easily extended to battery packs with the various number of battery cells connected in series. Experimental results show that the battery packs with 2 cells in series (2S), 4S, 6S, 8S and 10S can be heated efficiently and evenly with the temperature difference of the cells in the 10S battery pack within 2.5°C .

1.3. Article organization

The rest of this manuscript is organized as follows: Section 2 describes the design of the heating system based on LCC-S compensation topology and an electro-thermal battery model. Section 3 discusses the heating experiment and Section 4 concludes this study.

2. Heating system based on LCC-S compensation topology

2.1. Limitations of existing AC heating circuits

The purpose of the developed AC heating system is to preclude the need for an additional power system in EVs, relying on which the batteries can adjust the AC heating and charge-discharge strategies. Researchers have proposed a variety of on-board AC heating structures, which overcome the disadvantage of large AC power supply volume, but the self-oscillating structures are affected by battery power and may face a risk of failure at low SOC levels. In addition, different series and parallel battery packs may require additional or reduced electronic components to adjust the circuit structure for heating.

On the other hand, the secondary circuit of the wireless charging system is a natural high-frequency AC power supply affected by the primary high-frequency magnetic field, which can be easily extended to different battery packs, but it is only used to charge the battery after rectification.

Therefore, if the circuit structure of the wireless charging system is improved and the natural high-frequency AC power at the secondary side is fully utilized, the problem of AC power supply will be directly solved even for battery packs.

2.2. Design of proposed AC heating system

AC heating system has been designed based on a wireless energy transfer system, its structure is shown in Fig. 1, where U_{in} is a DC power source, which provides energy for the whole system. When DC passes through an inverter consisting of power MOSFETs $S_1 - S_4$, it is converted into high-frequency AC which is then applied to LCC, including the compensation inductor L_{f1} , the parallel compensation capacitor C_{f1} and the series compensation capacitor C_1 . L_1 and L_2 are the transmitting coil of the primary side and the receiving coil of the secondary side, respectively. M represents mutual inductance between coils. The diodes $D_1 - D_4$ are a rectifier that can convert high-frequency AC into DC. The switches $S_5 - S_6$ are always turned on and off at the same time. When $S_5 - S_6$ are turned on, the circuit works in heating mode. When they are turned off, the proposed circuit works in charging mode. U_b represents

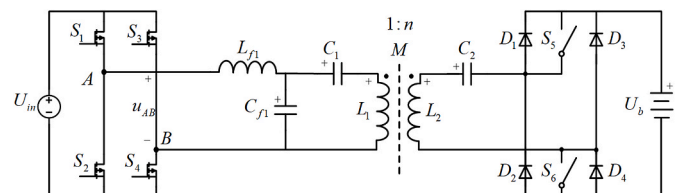


Fig. 1. AC heating system based on the wireless energy transfer system.

the voltage of the battery to be heated.

Given the high-order system shown in Fig. 1, this study focuses on the resonance characteristics of the topology at a constant frequency, and the system also realizes heating and charging functions at this high-frequency resonance point. The parameters of the circuit should meet [29,30]:

$$\begin{cases} \omega L_1 - \omega L_{f1} = \frac{1}{\omega C_1} \\ \omega L_{f1} - \frac{1}{\omega C_{f1}} = 0 \\ \omega L_2 - \frac{1}{\omega C_2} = 0 \end{cases} \quad (1)$$

where ω represents the angular frequency of the input voltage, which can also be expressed as: $\omega = 2\pi f$, and f represents the resonant frequency of the system [31].

In heating mode, the AC at the secondary side can be used to heat the battery. To simplify the analysis of the proposed circuit, the resistances of all capacitors are ignored [25], and then the equivalent circuit of the proposed heating system is shown in Fig. 2.

In this figure, R_{f1} and R_1 are the internal resistance of the inductor L_{f1} and L_1 , respectively, R_2 represents the internal resistance of the receiving coil L_2 . The part in the blue dotted box is the battery equivalent circuit model. In this model, there is the presence of both DC power and AC power which affect the current state of the circuit. According to the superposition principle, the circuit can be equivalent to Fig. 3.

Fig. 3(a) represents the equivalent circuit with only the AC power supply on the secondary side, and U_s can be expressed as:

$$\dot{U}_s = \dot{I}_s \left(R_2 + \frac{1}{j\omega C_2} + \frac{1}{j\omega C_D} \right) / \left(R_D + R_o \right) \quad (2)$$

Fig. 3 (b) represents the equivalent circuit with only the DC power supply on the secondary side. Due to the existence of the capacitor C_2 , we can obtain

$$\dot{I}_o = 0 \quad (3)$$

When the AC power supply and the DC power supply work together, we obtain the current of the secondary side

$$\dot{I}_2 = \dot{I}_s + \dot{I}_o = \dot{I}_s \quad (4)$$

Thus, it can be concluded that the DC voltage source does not affect the magnitude and phase of the secondary current, so the circuit of the system can be further equivalent to Fig. 4 for AC heating.

According to Kirchhoff's Voltage Law, we can get [28]:

$$\begin{cases} \dot{U}_1 = R_{f1}\dot{I}_{f1} + j\omega L_{f1}\dot{I}_{f1} + j\omega L_1\dot{I}_1 + \frac{1}{j\omega C_1}\dot{I}_1 + R_1\dot{I}_1 + j\omega M\dot{I}_2 \\ j\omega L_1\dot{I}_1 + \frac{1}{j\omega C_1}\dot{I}_1 + R_1\dot{I}_1 + j\omega M\dot{I}_2 = \frac{1}{j\omega C_{f1}}(\dot{I}_{f1} - \dot{I}_1) \\ \dot{U}_2 = R_2\dot{I}_2 + \frac{1}{j\omega C_2}\dot{I}_2 + j\omega M\dot{I}_1 \\ \dot{U}_2 = -R_o\dot{I}_2 - Z_D\dot{I}_2 \end{cases} \quad (5)$$

where $Z_D = \frac{1}{j\omega C_D} + R_D = \frac{R_D}{j\omega C_D R_D + 1}$ indicates the polarization impedance of the battery, U_1 is expressed as the voltage after the DC voltage U_{in} passes through the inverter, and the relationship with U_{in} can be expressed as [28]:

$$u_1(t) = \begin{cases} -U_{in} \left(nT < t < \frac{2n+1}{2}T \right) \\ U_{in} \left(\frac{2n+1}{2}T < t < (n+1)T \right) \end{cases} \quad (6)$$

According to (5), the input current and heating current can be solved separately:

$$\dot{I}_{f1} = \frac{\omega^2 C_{f1}^2 A}{\omega^2 C_{f1}^2 A R_{f1} + 1} \dot{U}_1 \quad (7)$$

$$\dot{I}_2 = -\frac{\omega^2 M C_{f1}}{Z + R_2 + (Z + R_2)\omega^2 C_{f1}^2 R_{f1} R_1 + \omega^4 C_{f1}^2 M^2 R_{f1}} \dot{U}_1 \quad (8)$$

where $Z = Z_D + R_o$ and $A = R_1 + \frac{\omega^2 M^2}{R_2 + Z}$. According to (6), when we ignore the loss of the inverter we can obtain the input power of the system as [27]:

$$P_{in} = U_1 I_{f1} \quad (9)$$

Since the battery generates heat under an AC excitation of a certain frequency, the electro-thermal model of the battery in the frequency domain is shown in Fig. 5, where U_{ocv} indicates open circuit voltage (OCV), which will be affected by battery power, U_B is the battery terminal voltage, $R_o(T)$ represents the ohmic resistance, $R_D(T)$ represents the electrochemical polarization resistance, and $C_D(T)$ is the electrochemical polarization capacitor.

3. Results and discussion

3.1. Experiment setup

A prototype of the heating system is built, which can provide the power of 500 W. In order to reduce the magnetic flux leakage of the secondary side which can improve the efficiency of the AC heating system, the coil of the secondary side is designed to be larger than the

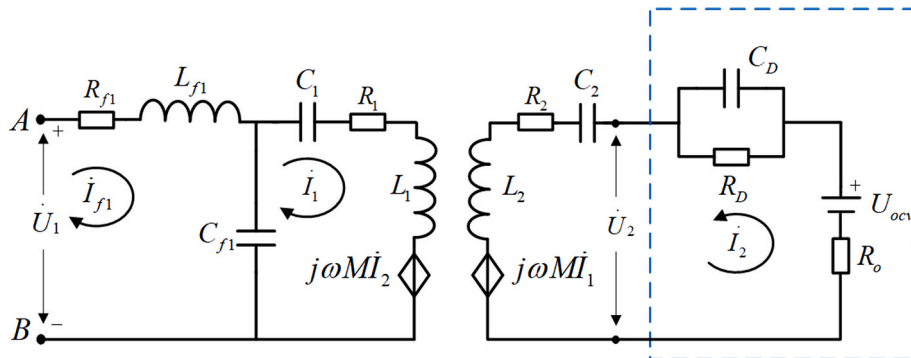


Fig. 2. Equivalent circuit referred to the proposed topology.

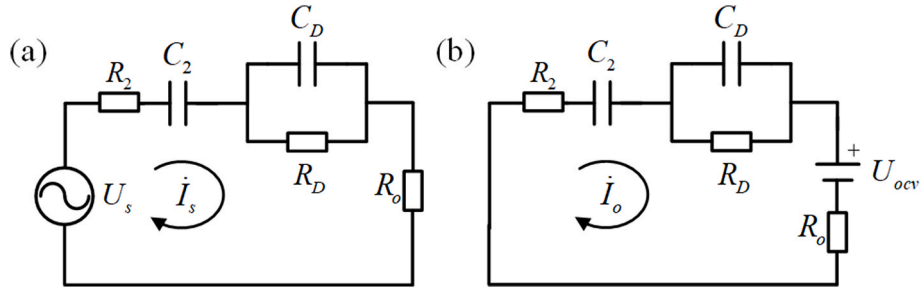


Fig. 3. Equivalent circuit of the secondary side: (a) Only AC power (b) Only DC power.

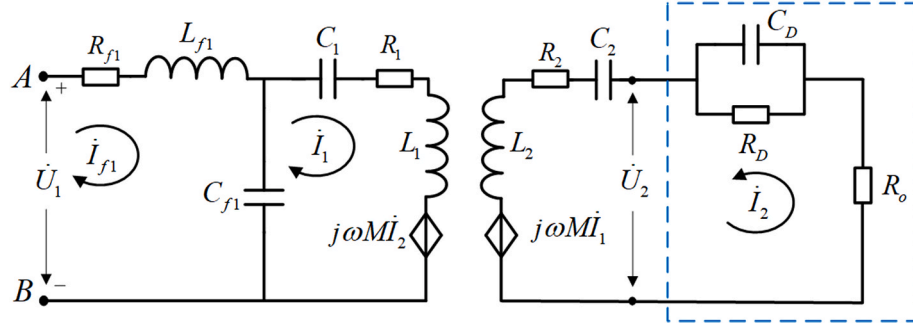


Fig. 4. Final equivalent circuit of the system.

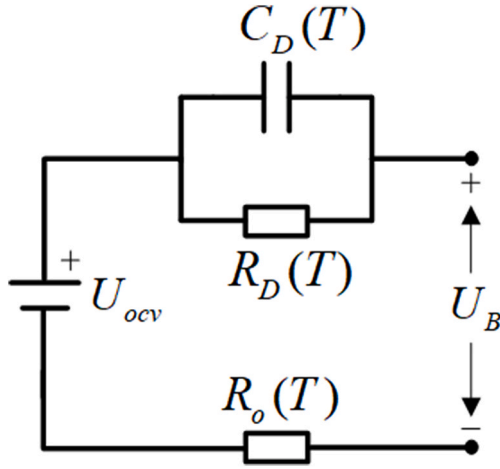


Fig. 5. Equivalent battery model.

primary side. The parameters of the designed circuit are shown in Table 1. During the experiments, the data collector (MEMORY HiLOGGER LR8450-01) with a K-type thermocouple can monitor and log temperature data in real time. And the oscilloscope (Tektronix MSO46) with current probe (TCP00300A) can be used to observe the current waveform. In order to ensure the low temperature environment and safety of the experiment, all experiments are carried out in an

Table 1
System design parameters.

Parameter	Design value	Measurement value
L_{f1}	13 μH	13.2 μH
L_1	20 μH	20.8 μH
L_2	28 μH	28.2 μH
C_{f1}	343 nF	343.5 nF
C_1	405 nF	405.2 nF
C_2	160 nF	160.4 nF

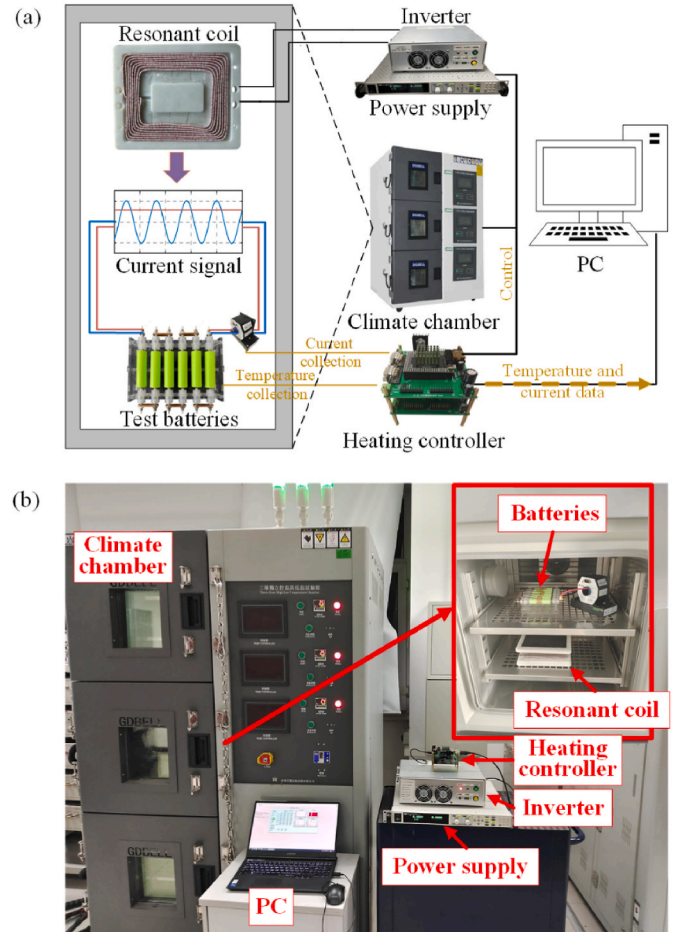


Fig. 6. Experimental equipment: (a) Frame diagram and (b) Physical diagram.

explosion-proof climate chamber (BE-8304-1000MT). The equipment used in the experiment is shown in Fig. 6.

Twelve brand-new 2600-mAh battery cells are selected for experiments. They are cylindrical 18,650 battery cells, and their specifications are listed in Table 2.

3.2. Heating current waveform verification

According to the theory in Section 2, the designed system can generate sinusoidal AC on the secondary side when the DC voltage is applied on the primary side. The amplitude of the secondary side AC will change when the input DC voltage changes, we can calculate the magnitude of the AC according to Eqs. (6) and (7). We choose one battery for pre-heating experiments to verify the effectiveness of the system to generate AC on the secondary side and model accuracy.

Fig. 7 shows the experimental results and corresponding calculations. In order to explore the performance of the system at low temperatures, the whole experiment was carried out in an environment of -20°C , and the battery has an SOC of 50%. As shown in Fig. 7(a), U_1 is the voltage input from the DC power supply to the primary side, the inverter generates a sinusoidal AC with a frequency of 85 kHz on the secondary side. The RMS value of the heating current is about 5.2 A (2C) when the amplitude of the input voltage is 36 V. Fig. 7(b) shows that, according to the calculation results, the effective value of the output heating current is proportional to the input voltage. The results also show that there is an approximately linear relationship between the DC input voltage and the effective value of the AC output heating current. Furthermore, the calculated currents agree well with the measured values, showing the validity of the Eq. (7).

3.3. Discussion of heating effect under different factors

(1) Different frequencies

Existing research on AC heating shows that AC frequency is an important factor affecting the heating rate [32]. However, the existing research focuses on low-frequency AC power due to the difficulty of increasing the frequency. In a small number of high-frequency heating studies, the AC excitation is not a standard sinusoidal current. In order to explore the impact of standard sinusoidal high-frequency current, we adjusted the resonant frequency according to the design principles outlined in Section 2, built the AC heating system with different frequencies and carried out AC heating experiments at different frequencies (20 kHz, 40 kHz, 60 kHz, 85 kHz, 100 kHz, and 150 kHz) while maintaining the same root mean square (RMS) value of the current. Fig. 8 shows the battery heating results. Before each heating experiment, the battery was placed in a thermal chamber at -20°C for 2 h to ensure that the battery is fully cooled. During the experiment, we keep the RMS AC at 7800 mA (3 C) and the SOC of the battery is 50%. Under sinusoidal AC excitation at a frequency of 20 kHz, it took about 380s for the battery to be heated from -20°C to 0°C . When the frequency increases to 40 kHz and 60 kHz, the heating durations are 268s and 184s, respectively. When the frequency further goes up to 85 kHz, the duration further reduces to about 145s. Surprisingly, it only took about 72s at 150 kHz. Although the heating duration varies for different frequencies, the

temperature rise of the battery is almost linear.

The experimental results show that increasing the frequency of the standard sinusoidal AC can effectively increase the heating rate. The frequency of the developed system is subject to the international standard of wireless charging [33], which has a natural high-frequency advantage and is simple and easy to obtain at a low cost. The same experimental results show that the battery can be effectively heated under the standard sinusoidal AC of 85 kHz. Compared with other non-standard high-frequency AC, the standard high-frequency sinusoidal AC has the advantage of a small peak value under the same RMS value, reducing the possibility of damage to the battery. These advantages make the designed structure more practical.

(2) Different battery SOCs

To explore the effect of different SOCs on the temperature rise of the battery, we carried out pre-heating experiments at the battery SOCs of 10%, 30%, 50%, 70% and 90%, respectively.

Fig. 9 shows the heating experiment results for one battery at different SOCs. At an ambient temperature of -20°C , we measure the heating current of the secondary side of the battery at different SOCs while keeping the input voltage constant. Then, we keep the RMS value of the heating current at 3C by adjusting the input voltage to heat the battery. As shown in Fig. 9(a), the standard deviation (Std) of the heating current to the battery at different SOCs is 0.0021, 0.0047 and 0.0043 A when the input voltage is at 10 V, 20 V and 30 V, respectively. Fig. 9(b) shows that the heating process varies with different battery SOCs, which has a great relationship with the impedance parameters of the battery, but due to the rapid heat generation of the AC excitation, the difference in the total heating time is not significant.

(3) Various current amplitudes

Another important factor affecting the AC heating rate is the current magnitude. Increasing the amplitude of the AC can effectively increase the heating rate [32]. To explore the effect of current amplitude on the heating rate at a frequency of 85 kHz, we carried out heating experiments at different RMS AC values of 2C, 2.5C and 3C and different insulation measures around a battery, respectively.

Fig. 10 shows the results of the temperature rise of the battery using different current amplitudes. In order to reduce the heat dissipation of the battery during the heating process, a thermal insulation layer is placed around the battery for comparison. Experimental results show that increasing AC amplitude can effectively increase the heating rate. When there are no insulation measures, the heating rate of the battery is about $3.1^{\circ}\text{C}/\text{min}$ given a current RMS amplitude of 2C. After increasing the current RMS value to 2.5C, the heating rate also increased to about $5.2^{\circ}\text{C}/\text{min}$. When the current RMS value increased to 3C, the heating rate was twice as high as that of 2C, reaching $8.2^{\circ}\text{C}/\text{min}$. When there is an insulation measure, the heating rate has also been improved to a certain extent. The improvement in temperature rise rate is most obvious under the AC RMS value of 3C, it is because the heat dissipation of the battery does not change under the current of different amplitudes, but the heat generation is higher when the current amplitude is larger.

Therefore, under the AC excitation of 85 kHz, increasing the AC heating current amplitude and taking certain insulation measures can effectively increase the heating rate. The heating system designed in this paper can directly increase or decrease the output heating current of the corresponding secondary coil by increasing or decreasing the input voltage to adjust the heating rate.

3.4. Heating for battery packs

Existing AC heating studies are generally for a single cell [34]. Owing to the limitations of the adopted heating systems, few of them can be extended to battery packs. In reality, battery packs are required in EVs.

Table 2
Specifications of the tested batteries.

Parameter	Value
Battery type	LR1865SK
Nominal capacity	2600 mAh
Upper voltage	4.2 V
Lower voltage	2.7 V
Maximum discharge current	7800 mA (3C)
Battery mass	48 g

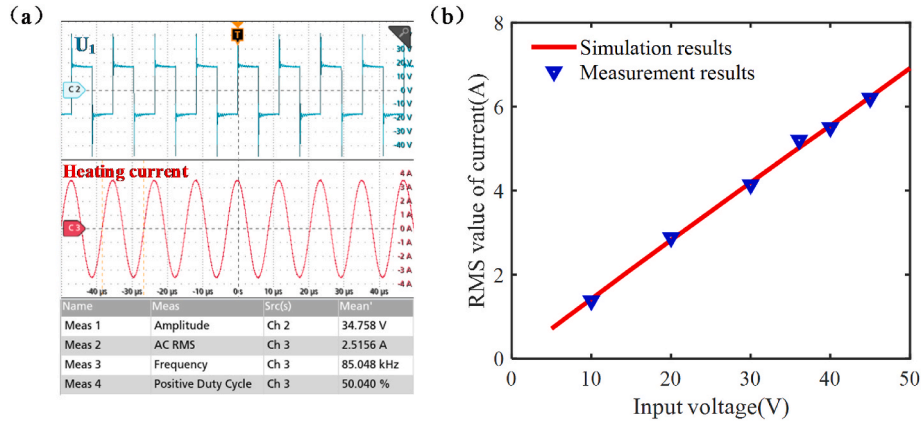


Fig. 7. Experimental and calculation results of output heating current (a) Secondary current in the experiment (b) Simulation and measurement results of output heating current.

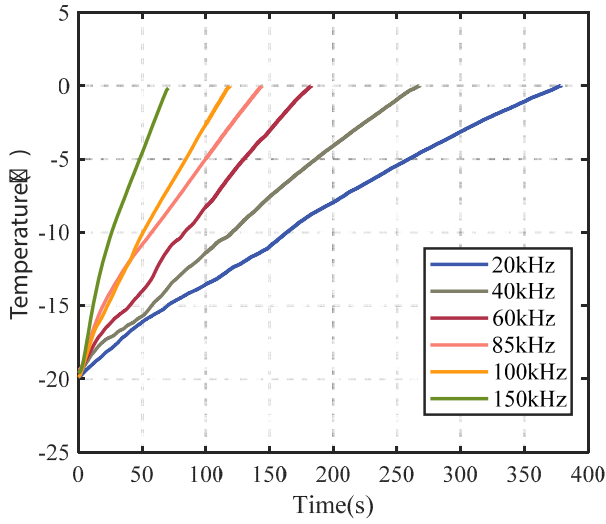


Fig. 8. Results for battery at 50% SOC with different frequencies while same RMS current of 7800 mA (3C).

To apply AC heating to a battery pack, some existing AC heating structure needs additional components, and the heating rate are subject to battery SOC [32]. Thanks to the scalability of the developed heating

system in this study, it can be directly applied to battery packs without additional costs. To investigate the effectiveness of the proposed heating method for battery packs, we carried out heating experiments of the battery packs with two cells in series (2S), 4S, 6S, 8S and 10S for the given AC.

Fig. 11 shows the heating results of the battery packs. All the

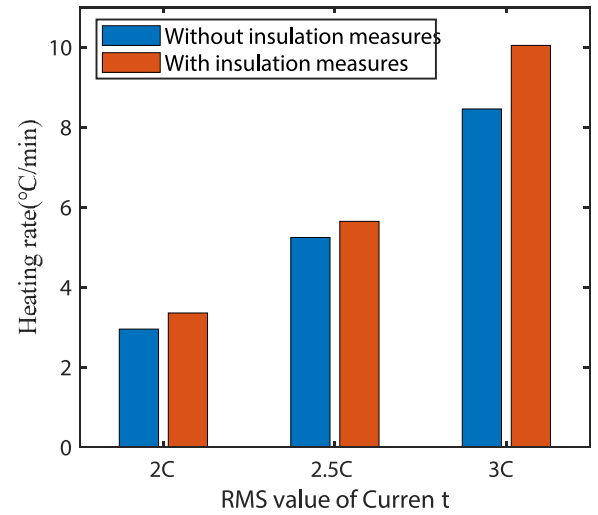


Fig. 10. Battery heating rate using different RMS values.

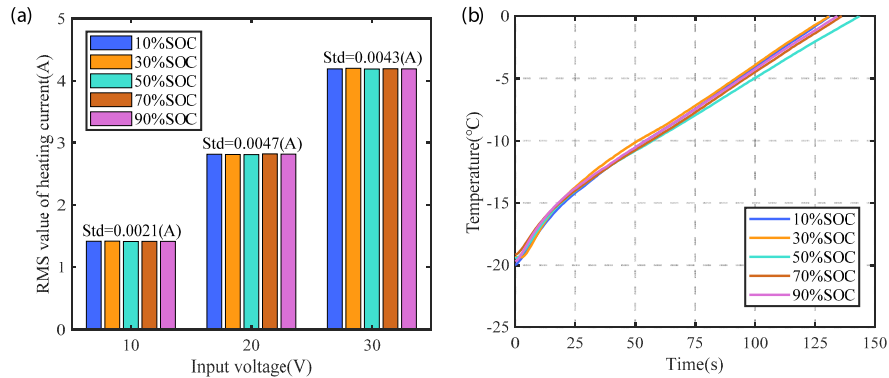


Fig. 9. Results for a battery at different SOCs. (a) Current vs. voltage; (b) Temperature rise curves.

experiments of the battery packs were carried out at an ambient temperature of -20°C , and the SOC of each cell in the battery pack was 50%. As shown in Fig. 11(a), the proposed heating system can generate sinusoidal AC to the battery packs with different numbers of cells in series when different input DC voltages are applied, and the RMS value of the output AC will increase proportionally with the increase of the input DC voltage. The RMS value of the heating current of the 2S battery pack is about 6.3 A when the input voltage is 10 V, and the RMS values for 4S and 6S battery packs are 3.8 A and 2.0 A, respectively, but the heating current is only 1.2 A for the 10S battery pack. It can be seen that the heating current will decrease with the increase of the number of the cells in series for a battery pack under the same input. The results in Fig. 11(b) indicate that a 10S battery pack can be effectively heated from -20°C to 0°C in about 140s. The difference between the highest cell temperature and the lowest cell temperature in the battery pack is within 2.5°C , indicating that the designed heating system can heat the cells consistently in the battery pack.

Fig. 12 plots that the battery temperature rises. Fig. 12(a) shows that the proposed heating system can heat the batteries evenly, and the temperature difference between the middle and the side cell is 0.7°C . Fig. 12(b) indicates that the temperature difference of all cells is within 1°C . It also shows that the surface temperature of each cell is almost the same during the complete heating process, which indicates that the battery pack is heated efficiently and uniformly.

3.5. Comparison with different heating methods

Finally, to evaluate the performance of the proposed heating system, a systematic comparison is made with the widely adopted four AC heating methods. The comparison results include power supply, current waveform, heating speed, adaptation to the performance of the battery pack, the number of components, system efficiency and cost, they are shown in Table 3.

By applying an external AC power source (Kikusui bipolar supply), the low-frequency AC heating method [23] heats the battery with 10–60 Hz sinusoidal AC, the battery can reach an average temperature-rise rate of $2.41^{\circ}\text{C}/\text{min}$ with very little damage and maintain high efficiency. Although this method has also been tested on battery packs, all its experiments rely on complex, bulky and expensive equipment, which is not suitable for EV applications.

By changing the structure of the battery, the method in Ref. [14] can achieve self-heating within a dozen seconds by controlling the heating switch. Self-heating batteries have the advantages of high heating speed, low cost and high efficiency. Besides, it works well with battery packs. However, self-heating batteries still face many problems due to structural changes before large-scale application.

To overcome the limitation of the external AC heating power supply,

the onboard heater in Ref. [35] generates AC by means of self-oscillation, which can realize EV applications. This buck-boost structure produces a triangular waveform AC. Although it has characteristics of high efficiency and high heating speed, this waveform is more harmful to the battery than the sinusoidal waveform at the same value of RMS [32]. SW heater was improved on the basis of the buck-boost structure [17], which can generate non-standard sinusoidal AC by self-oscillation to heat the battery. They both are suitable for battery packs, but they require a certain number of additional components, which may increase the complexity of the control.

By comparison, the proposed heating system generates standard sinusoidal AC based on the wireless energy transfer system to heat the battery cells or battery packs. Standard sinusoidal AC has smaller peaks than other asymmetrical AC curves at the same RMS value, which means less damage to the battery. In addition to the advantages of high heating speed, low cost and high efficiency, this method can be easily applied to battery packs with any number of battery cells in series, making it more suitable for EVs.

3.6. Heating and battery aging

Preheating battery at low temperature may cause its aging process to accelerate, so it is important to discuss the effect of the proposed heating method on aging. In order to verify the impact of the proposed method on battery health, we have conducted 200 cycles continuous heating experiments. During the experiment, the batteries were kept at 50% SOC and heated from -20°C to 0°C with a heating current of 3C RMS value. The charge and discharge capacity of the fresh battery before heating experiment are 2.528Ah and 2.5311Ah respectively. After 200 cycles of heating, the charge and discharge capacity of the battery become 2.543Ah and 2.542Ah respectively. The battery has not been significantly degraded after the 200 heating cycles. Fig. 13 shows the battery capacity calibration test of the battery. It can be found that the 200-time continuous heating did not cause significant aging to the battery, and this process also includes the normal aging of the battery itself.

4. Conclusion

This study proposes an AC heating method for lithium-ion batteries at low temperatures by using a wireless energy transfer system. It has the advantages of high efficiency, low cost and easy expansion to battery packs. Through a detailed analysis of the experimental results of battery cells and battery packs, some conclusions are drawn:

- (1) The developed heating method based on the wireless energy transfer system can generate sinusoidal AC with a frequency of 85 kHz, which can efficiently heat the battery with a natural high-

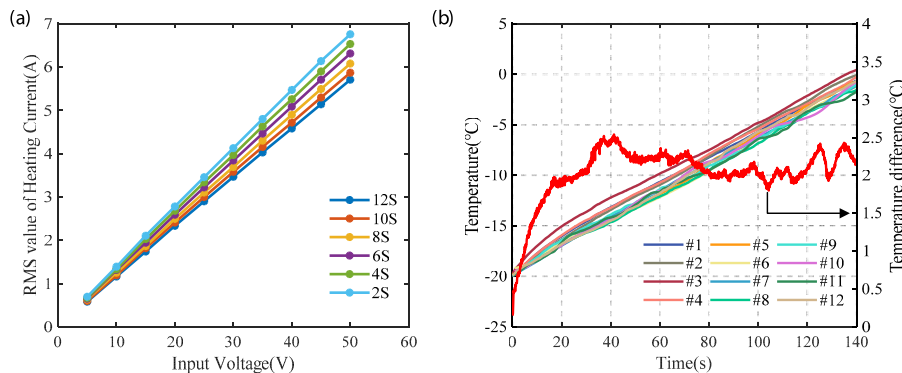


Fig. 11. Results for battery pack. (a) Heating current vs. input voltage with different numbers of cells in series; (b) Temperature rise of cells #1 to #12 and difference between the maximum and minimum temperature.

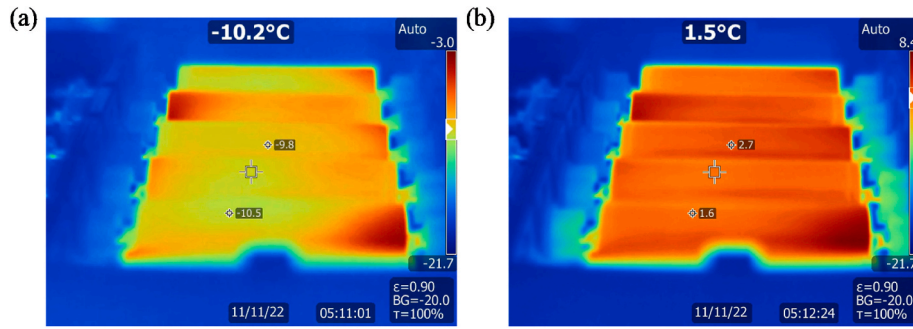


Fig. 12. Temperature rise results of 5 series of batteries during heating. (a) Temperature distribution of battery pack when heated to -10 °C (b) Temperature distribution of battery pack at the end of heating.

Table 3

A comparison of different heating methods.

Heating methods	Power supply	Current waveform	Speed	Expandability (battery packs)	Component number	Efficiency	Cost
Low-frequency [23]	External	Sinusoidal	Slow	High	Large	Low	High
Self-heating heater [14]	Battery	DC	High	High	Little	High	Low
Onboard battery [35]	Battery	Triangular	High	Medium	Medium	High	Medium
SW heater [17]	Battery	Sinusoidal	High	Medium	Medium	High	Medium
Proposed method	External	Sinusoidal	High	High	Little	High	Low

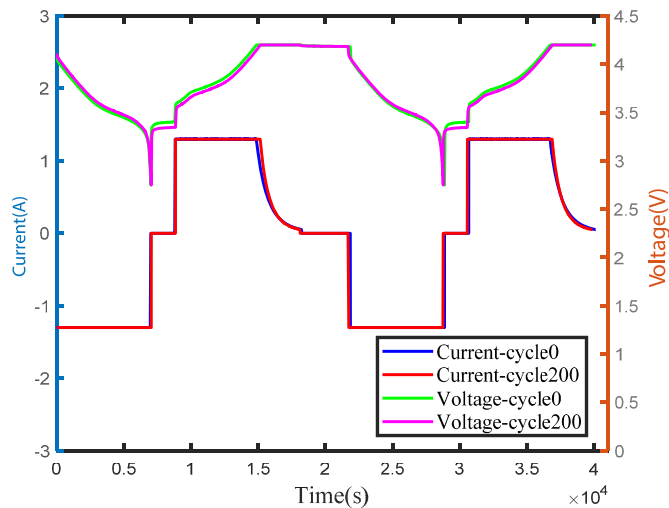


Fig. 13. Capacity calibration before and after continuous heating test.

frequency advantage. Without the need for an external AC power supply, the designed system is practical in EV applications.

- (2) By changing the input voltage, the proposed heating system can heat the battery at different rates under different SOC. According to the actual temperature, battery status and working requirements, the heating strategy can be adjusted flexibly, which is more conducive to reducing the possible damage to the battery.
- (3) The proposed heating system can be easily extended to the battery packs which consist of different numbers of battery cells in series. The validation results show that the battery packs can be heated evenly and efficiently by adjusting the input voltage, however, increasing the number of series cells in the battery pack under the same input will reduce the heating current.

In summary, the system proposed in this paper can heat the battery with a high speed. However, there are not many EVs equipped with wireless charging systems, and efficient wireless charging systems are

still in the development stage. In the future, we will further explore how to apply the proposed heating in other conditions and improve the heating efficiency of high-frequency AC to battery cells and battery packs. Besides, the impact of different current amplitude on battery life is also worthy of exploration.

CRediT authorship contribution statement

Rui Xiong: Investigation, Methodology, Conceptualization, Writing – review & editing, Supervision. **Kui Zhang:** Methodology, Software, Writing – original draft. **SiYu Qu:** data, Software. **Jinpeng Tian:** Writing – review & editing. **Weixiang Shen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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